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REVIEW ARTICLE

The potential role of retinal slip in National Football League (NFL) officiating and its effect on on-field decision making: mechanisms and countermeasures

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Visual processing involves numerous neuro-ophthalmic pathways that enable humans to perceive and interact with the world. These visual pathways can negatively impact visual perception and decision-making. The Vestibulo-Ocular Reflex (VOR) is a compensatory mechanism that stabilizes gaze on a target during head movement, such as running or turning. However, inadequate VOR stabilization can lead to retinal slip, reducing visual acuity. In the National Football League (NFL), officials require exceptional visual training and performance. NFL officiating utilize rapid visual processing, strategic movements, and precise judgement to make accurate penalty calls in real-time. Retinal slip can adversely affect play calling and penalty assessment by NFL officials. Despite these challenges, NFL officials achieve high precision and accuracy in up to 98.9% of plays. To maintain such high standards, the NFL has implemented a standardized curriculum to mitigate the effects of VOR and retinal slip. This manuscript outlines the basics of VOR function and explores countermeasures to minimize retinal slip in NFL officiating. We believe that these scientific endeavours can improve both penalty assessment accuracy and precision. The use of these general principles might also be potentially applicable to other sports.

Eye; https://doi.org/10.1038/s41433-024-03533-z

INTRODUCTION

Visual processing involves numerous neuro-ophthalmic pathways that enable humans to perceive and interact with the world. The interaction and activation of these visual pathways can sometimes negatively impact our perception of the visual environment which can in turn affect decision making. The Vestibulo-Ocular Reflex (VOR) is a compensatory movement that stabilizes our gaze on a target during head movement (e.g., running or turning our head). However, if the VOR does not adequately stabilize the image on the retina, a retinal slip occurs, reducing visual acuity. In the National Football League (NFL), game officials require a high level of visual training and performance including rapid visual processing, strategic and timely head and body movements, as well as judgement and experience in order to make a precise and accurate NFL penalty call in real time. Retinal slip can affect both play calling and penalty assessment by NFL game officials.

NFL officials, who must perform at the highest level under high stress and high oversight (e.g., instant replay) conditions. Despite these demands, NFL game officials are able to achieve high levels of precision and accuracy in up to 98.9% of total plays [1]. In order to continue to perform at such high levels, the NFL has implemented a standardized curriculum and teaches specific principles to mitigate the effects of VOR and retinal slip on game official performance. We previously described the neuro-ophthalmology curriculum for NFL game officials [2]. In this manuscript we describe the basics of VOR function and the possible countermeasures to minimize retinal slip for NFL officiating. We believe that these scientific endeavours can improve both penalty assessment accuracy and precision. The use of these general principles might also be potentially applicable to other sports.

VOR AND RETINAL SLIP

Physiology

The human eye's intricate movements are controlled by six extraocular muscles (e.g., superior rectus, inferior rectus, medial rectus, lateral rectus, superior oblique, and inferior oblique muscles), working together to direct gaze and focus light onto the centre of the retina (i.e., the fovea). These extraocular muscles

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Received: 1 June 2024 Revised: 20 November 2024 Accepted: 29 November 2024
Published online: 06 December 2024

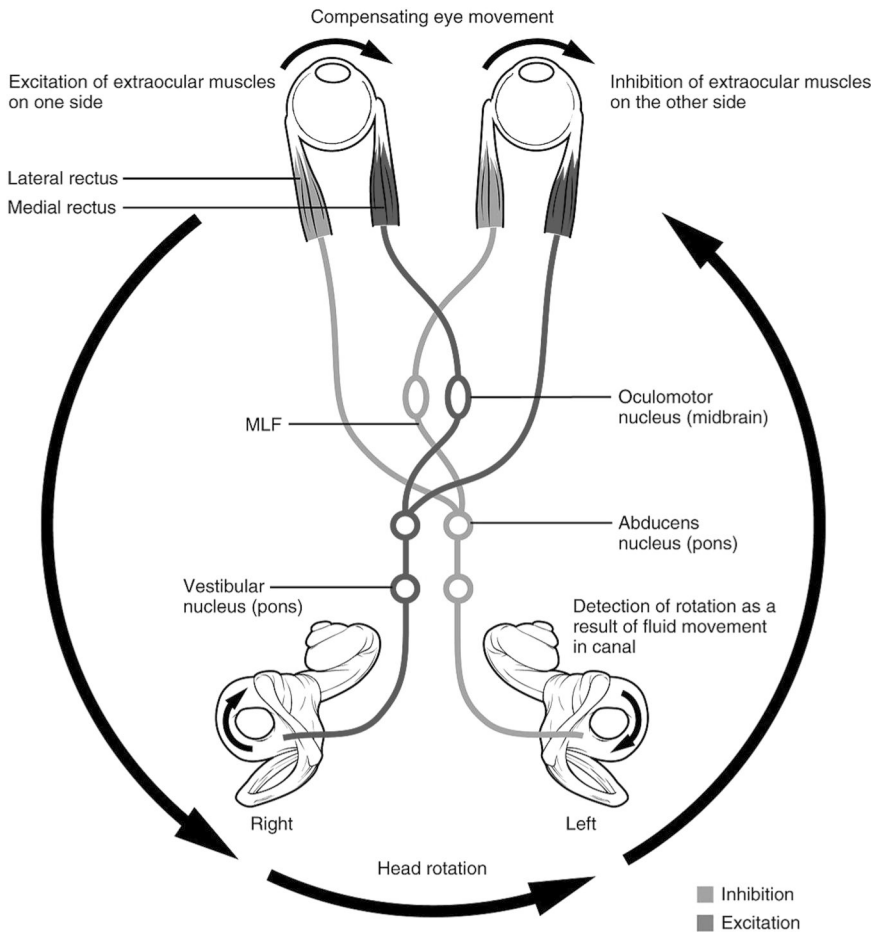

Fig. 1 Basic components of the VOR in the horizontal plane. Reprinted with permission from Wikimedia Commons, 1608 Vestibulo-Ocular Reflex-02, licensed under Creative Commons Attribution-Share Alike 3.0 Unported.

are under the control of three cranial nerves (CN III, IV, and VI) and create basic eye motions such as upward, downward, and horizontal conjugate gaze. Basic eye movements are initiated by signals originating in the cortex for prompting extraocular muscle contraction, thereby shifting the eye's position. However, more intricate tasks, like tracking a moving object or maintaining visual stability during motion, demand more sophisticated neural circuitry to ensure the foveal capture of a clear image. Some of these forms of eye movements include saccades, smooth pursuit, and vestibulo-ocular movements. Saccades are rapid eye movements from one focal point to another and are utilized on the field to rapidly changed focus between players on different points of the field [3]. Smooth pursuit is slower movements that allow for fixation on a focal point as it moves, such as watching the path of the football in the air [4]. Another mechanism facilitating visual acuity during movement is the vestibulo-ocular reflex (VOR). The VOR achieves image stability during motion through equal and opposite compensatory eye movements, leveraging input from peripheral sensors within the vestibular system [5]. Similar to the VOR, there is an the optokinetic response (OKR), that provides compensatory movements in the same direction as the visual when there is a large motion in part of the visual field such as looking at a tree in a moving vehicle [6]. The VOR consists of peripheral sensors, central processing mechanisms, and motor outputs. The interplay between the horizontal motion components is depicted in Fig. 1, illustrating the dynamic nature of the VOR's functioning [7].

The semicircular canals consist of six canals (anterior, posterior, lateral) detecting angular velocity across multiple axes. The utricle and saccule are otolith organs that also contribute to the VOR through activation of hair cells that relay positional information regarding head translation and static head tilt [8]. Fig. 2.

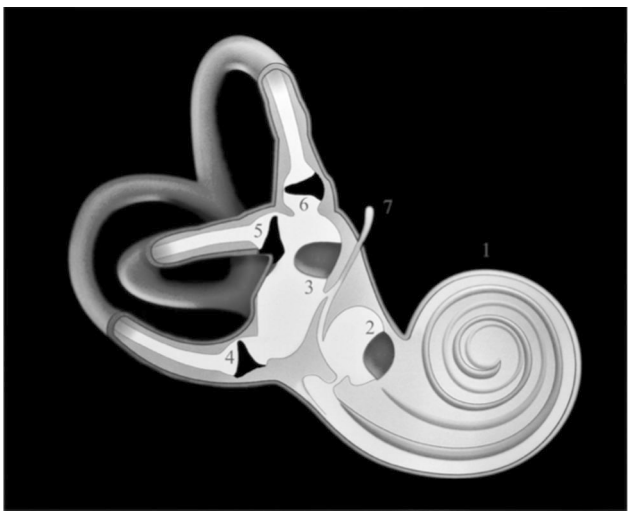

Fig. 2 Semicircular Canal anatomy: 1-Cochlea 2-Saccule 3-Utricle 4-Posterior ductus ampulla 5-External ductus ampulla 6-Superior ducutus ampulla 7- Ducutus endolymphaticus Reprinted with permission from Wikimedia Commons, Oreille Interne, licensed under Creative Commons Attribution-Share Alike 3.0 Unported.

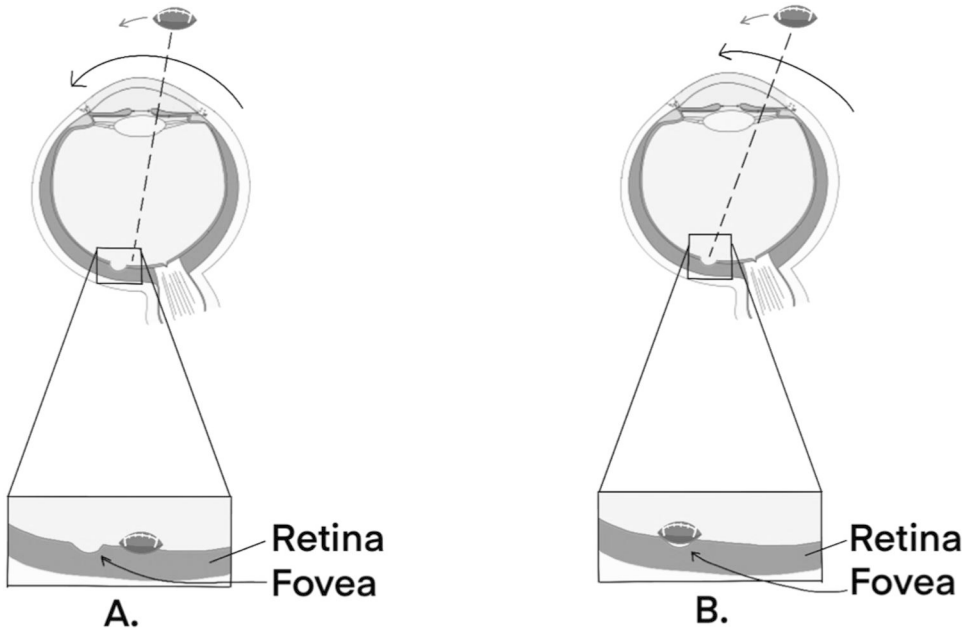

Fig. 3 Visual Representation of Retinal Slip. A Retinal slip causing a blurry image. **B** Corrected retinal slip with the image properly focused on the retina. Eye diagram adapted with permission from Wikimedia Commons, Pereru, under Creative Commons Attribution-Share Alike 4.0 International license. https://commons.wikimedia.org/wiki/File:Simple_diagram_of_the_human_eye.png.

Horizontal positional information is relayed through the abducens nucleus in the pons to the oculomotor nucleus in the midbrain via the medial longitudinal fasciculus (MLF) [9]. These signals trigger equal and opposite horizontal eye movements, stabilizing the moving image on the fovea. Likewise, the vestibular nuclei combines input from the semicircular canals and otolith organs, which is further relayed to the oculomotor and trochlear nuclei within the vertical gaze centres in the thalamomesencephalic region to control vertical gaze [10].

The velocity of head rotation, in any axis, is the primary stimulus to engage the VOR. Studies on horizontal motion found that head movements ≥2 Hz activate the VOR, prompting compensatory ocular movement aimed at stabilizing the eyes on a fixed focal point [11]. For accurate perception, the image must remain within 0.5 degrees of the fovea, with declining visual acuity observed as the focus deviates from this central point [12]. Because the VOR is reacting to head movement, it has a feedback mechanism that allows for changes in velocity if the image is not focused appropriately. If the VOR fail to generate the correct compensatory movement, the image will not remain focused on the retina, resulting in a blurred or unstable image [13]. This decrease in visual acuity during the VOR is related to the degree of movement across the retina, known as retinal slip. Figure 3. Figure 3 illustrates how retinal slip can result in an image not being properly focused on the retina.

The OKR, VOR and smooth pursuit are all neural process that work to reduce retinal slip and create clear images on the retina [14]. Our visual acuity has been shown to decrease when retinal slip reaches a velocity of ~3°/s [15]. While our eyes are experiencing a decrease in visual acuity, it is frequently not perceived consciously because higher level processing can suppress the retinal slip [16]. Because at this level of higher order cortical processing, the individual themselves may be unable to perceive what was actually seen or missed. Although this higher level processing is good at "filling in the gaps" of our visual field and visual acuity, it can not fully account for even minor retinal slip. Greater deficits in the VOR function can induce motion sickness, dizziness, and oscillopsia associated with head motion [17].

Although the VOR does an excellent job at facilitating the observation larger objects, such as tracking players in motion, it impairs our perception of finer details, such as focusing on the goal line [12]. The reduction in high spatial frequency sensitivity during VOR activation complicates the discernment of details when in motion. This phenomenon is demonstrable in practical scenarios, like viewing text on a screen while moving one's head back and forth. Although the ability to maintain focus on the text at low levels of motion is relatively good, as the degree of retinal slip increases the clarity of images and spatial resolution declines.

Research on walking speed has corroborated these findings of retinal slip with visual acuity degrading with faster walking speeds [18]. Walking indirectly activates the VOR through vertical head movement and slight nodding during motion. In addition, a separate study quantified the reduction in visual acuity during head movement, showing a logarithmic decline beyond 5°/s [19]. This decline in visual acuity is a challenge for NFL officials, who are tasked with making critical calls while in body motion or head rotation during game play. Thus, NFL game officials must balance the needs to track the ball and player trajectory; to monitor players' movements; and to maintain a safe distance from the action against the beneficial and minimizing effects on retinal slip of remaining stationary and refraining from head rotation. Game officials must be trained on developing optimal viewing and movement strategies to mitigate retinal slip and to ensure accurate decision-making on the field. While our preliminary research highlights the effects of retinal slip in NFL officiating, further studies are necessary to fully quantify the resulting changes in visual acuity caused by this mechanism.

VOR AND RETINAL SLIP

Applications in football officiating

Visual performance during NFL game play affects both the athletes and game officials. The ability to track objects and coordinate physical movement in real time are essential elements of play on both sides of the ball. Research has quantified this fundamental relationship, demonstrating that athletes exhibit enhanced dynamic visual acuity (DVA) compared to non-athletes [19]. Visual

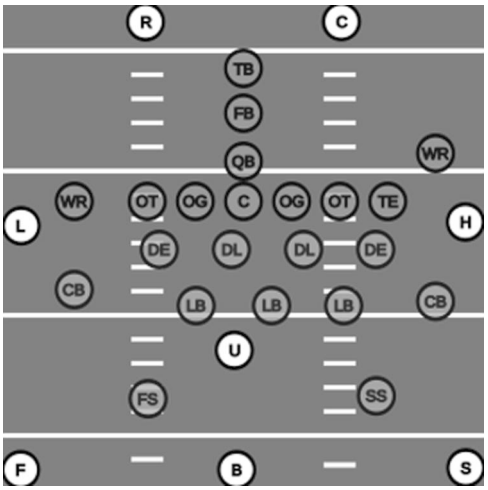

Fig. 4 General Football Officials Positioning: The Referee (R), Umpire (U), Head Linesman (H), Line Judge (L), Field Judge (F), Side Judge (S), Back Judge (B). Reprinted with permission from Wikimedia Commons, Zzyzx11, under Creative Commons Attribution-Share Alike 4.0 International license.

reaction time has also been shown to be increased in some athletes compared to non-athletes [20]. Although athletes display an increased DVA and visual acuity, the specific visual demands vary greatly across different sports. For instance, some sports like track and field may necessitate broader visual awareness over fine visual processing, whereas sports such as shooting rely heavily on high visual acuity for precise targeting. Notably, studies conducted at the Olympic level have revealed significant disparities in visual performance among different sports, with boxers and track and field athletes exhibiting notably lower visual acuity levels [21].

The position a player occupies can significantly influence their dynamic visual acuity and overall performance. For instance, a study conducted on soccer players revealed that goalkeepers exhibited notably higher DVA levels compared to forwards. Additionally, a greater proportion of goalkeepers were found to utilize optical correction for best corrected vision, underscoring the importance of visual acuity precision in their role on the field [22]. The VOR also adapts to meet the demands of specific athletic activities. Athletes in ball sports like football typically exhibit superior VOR compared to those in combat sports such as taekwondo, wrestling, and judo [23]. This difference arises from the coordination required by ball sport athletes to synchronize their eye and head movements when catching or hitting a ball, creating a high degree of control and refinement in their VOR.

Just as elite athletes rely on optimal visual performance to excel in their respective sports, elite sports officials must also possess a high level of visual acuity to effectively assess plays and make accurate decisions. Research on soccer officials has revealed a significant link between visual performance and on-field officiating performance. Studies focusing on soccer officials have demonstrated superior stereoacuity and visual memory are associated with higher rankings in on-field performance among officials [24]. Although research on visual performance among sports officials has not been as extensive as that for athletes, recent studies, particularly those conducted in collaboration with the NFL, highlight a growing interest in this area [2].

While there is no quantitative research specifically targeting the VOR in NFL officials, the dynamic movements and rapid decision-making inherent in their role engage the VOR and influence their calls. NFL football officials are responsible for overseeing 22 players on a sprawling 57,600 square foot field. These officials, many of whom must navigate in real time on the field of play must focus on their designated responsibilities and manoeuvre strategically to ensure a clear and unobstructed line of sight as well as avoiding interference with the players and the flow of game play. Figure 4 depicts the general positioning in American Football of various officials, white dots.

This is no easy task, requiring a high level of knowledge and physical agility. NFL officials are organized into seven distinct positions, each with specific field placements and play-calling duties. Additionally, each official must move to effectively evaluate plays and their movements vary based on the type of play unfolding. For instance, the down judge is stationed at the line of scrimmage and focuses on detecting offside and encroachment during the snap. Once the play begins, this official must shift their attention to the point of play (i.e., the play mesh point) to determine if a player has stepped out of bounds during run plays, and to assesses whether a receiver is in or out of bounds during a pass play. In each scenario, officials must swiftly position themselves and acquire a clear visual target to accurately assess the unfolding play.

The VOR predominantly engages during head rotation and vertical head movement, particularly during running activities. For NFL officials, both of these motions may be essential for obtaining an optimal visual field which increases with increased distance. Increased distance from the play however reduces central visual acuity and motion of the head and/or body may trigger the VOR leading to unwanted retinal slip. Plays demanding significant head movement often occur when officials are running toward the action (VOR) or when tracking the football during a pass play to confirm a catch (completion or incompletion) and to determine if a player is in or out of bounds. As officials pivot their heads to observe the play, their eyes can experience a delay of up to 10 milliseconds behind the head movement, resulting in retinal slip and a temporary disruption in stabilizing the image [15]. As mentioned previously, the higher cortical processing centres in the brain suppress the retinal slip and anecdotally game officials often insist that they did not miss or perceive elements of game play even after watching on replay.

The activation of the VOR and restricted visual field experienced by on-field officials is not seen by spectators watching replays or television viewers at home. This disconnect contributes to the perception by disgruntled fans and opposing teams that penalty judgements made by officials on the field are incorrect due to incompetence or inattention when in reality, retinal slip may be the actual root cause for missed or mistaken calls.

Despite these limitations, experienced game officials can be trained to minimize the pitfalls of VOR and retinal slip. In fact, each game official typically works 15 regular-season games and monitors over 2100 plays, averaging approximately 153 plays per game. Remarkably, their precision remains consistently high, with an accuracy rate up to 98.9% [1].

SOLUTIONS AND FUTURE DIRECTIONS

With 17.9 million viewers per game, each NFL penalty call (or non-call) is a high pressure and high stakes game decision [25]. Because of this intense pressure the NFL sets the highest level of excellence for officials and has extensive training and evaluations to improve play calling. Each official must participate in two preseason clinics and weekly evaluations throughout the season to ensure officials are making the right call. Both before the season and throughout, officials spend time breaking down video to review previous calls made. The NFL has also started assessing whether a neuro-ophthalmic curriculum could improve precision and accuracy of NFL play-calling. A study on this curriculum found a significant improvement in neuro-ophthalmic knowledge and a greater likelihood of using NOP terms following the neuro-ophthalmology curriculum [26].

Currently the NFL has applied neuro-ophthalmology principles such as VOR to their “Move with a purpose” curriculum, which

integrates specific strategies aimed at maximizing visual acuity. This curriculum revolves around three key questions: Do I have an unobstructed view of the play? Do I need to reposition myself to obtain a clear view? And if so, how can I do so with minimal disruption to my visual acuity [27]? The focus of the third question centres on mitigating the visual disturbances induced by the VOR and minimizing its effects.

To minimize the impact of the VOR, two primary movement adjustments are utilized. Firstly, officials are instructed to avoid excessive motion (e.g., running with the play or not allowing the player to "beat the official to the goal line") or rapid head movements that may result in visual blurring caused by retinal slip. For instance, officials are encouraged to maintain their line of sight rather than rushing towards the play, thus preserving visual acuity. Secondly, officials employ a technique known as "breaking down," where they align their head with the play and pivot on one foot to minimize rapid head movements and prevent the activation of the VOR.

Further initiatives aimed at enhancing the visual quality of officials and improving play-calling accuracy hold significant promise, particularly through advancements in VOR. Given that the VOR is an adaptable system, it possesses the capacity for training and optimization. While performance characteristics with VOR adaptation training have not yet been extensively studied in relation to sports officials, prior findings suggest the potential for enhancing visual performance through VOR adaptation. Visual blur induced by retinal slip serves as a potential stimulus for driving VOR adaptation. Previous studies have demonstrated that induced retinal slip, achieved through lenses/prisms or moving displays, can effectively drive VOR adaptation [13, 28]. Additionally, research indicates that incremental VOR adaptation, which gradually increases retinal slip, leads to faster adaptation rates [29, 30].

For NFL officiating, VOR training can involve any method that enhances officials' understanding of how the VOR influences their vision, such as the existing neuro-ophthalmology curriculum, or changing the adaptation of the VOR using tools such as lenses, prisms, or virtual reality. One promising avenue for implementing VOR training is through extended reality (XR) platforms [31]. XR training has had substantial growth across various domains, including medicine and sports performance [32, 33]. Specifically, in ophthalmology, XR has been utilized for surgical simulation and functional patient assessment [34]. Recent efforts with NFL officials have explored the integration of neuro-ophthalmology principles into XR-based interactive training experiences. These simulations would provide officials with a realistic environment where they can apply neuro-ophthalmic principles [34]. Incorporating VOR training into such simulations can help underscore the adverse effects of rapid movement on visual performance. For instance, notifications can be triggered for officials when their head rotation reaches speeds that activate the VOR, thereby enhancing their understanding of the optimal range for movement while maximizing visual acuity.

CONCLUSION

The VOR compensates for head or body movements, allowing for steady vision during movement. However, when the eyes cannot fully compensate, a retinal slip occurs, decreasing visual acuity. The VOR's ability to adapt can minimize this loss of visual acuity during head movements, which can be advantageous to NFL officials, who frequently engage the VOR while moving to assess plays. This activation may lead to a decrease in visual acuity, potentially affecting play calls. The NFL's neuro-ophthalmology curriculum and XR programme are strategies aimed at leveraging principles like the VOR to improve play calling. Future developments in VOR training can focus on educating NFL officials when the VOR is engaged and creating specific training programmes to optimize their VOR function, ultimately enhancing their performance and decision-making accuracy.

DATA AVAILABILITY

No new data were created or analysed in this study. Data sharing is not applicable to this article.

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AUTHOR CONTRIBUTIONS

Conceptualization, JO and AGL, investigation, JO, NP, HM, and AGL; writing—original draft preparation, JO, NP, and HM, writing—review and editing, JO, NP, HM, NC, NZ, VL, AZ, NL, AT, EW, PS, and AGL; supervision, AGL; project administration, NP; All authors have read and agreed to the published version of the manuscript.

COMPETING INTERESTS

AGL serves as a consultant for the National Football League (NFL). Dr. Lee is on the Speakers' Bureau for Amgen and Alexion and is on the Advisory Board for Viridian and Stoke and serves as a consultant for Bristol Myers Squibb and Astrazeneca. Dr. Lee serves on the Editorial board of *Eye*.

ADDITIONAL INFORMATION

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41433-024-03533-z.

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